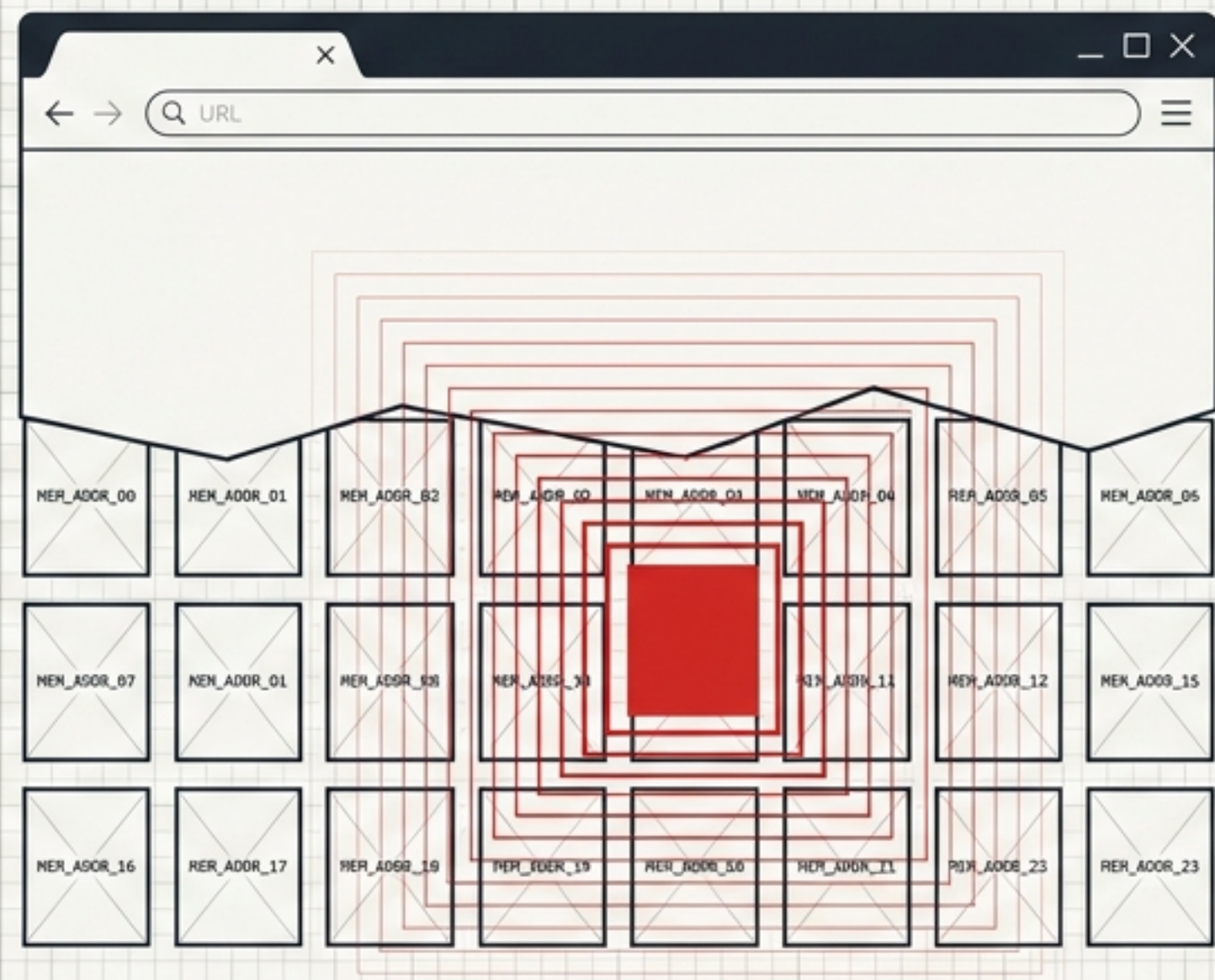


# Chrome Dawn WebGPU Memory Disclosure

## CVE-2026-11067 Vulnerability Analysis & Remediation Briefing



[Threat Intel]

[Severity: Medium (6.5)]

[Root Cause: CWE-457]



# Executive Threat Dashboard

Base Metrics Scorecard

6.5

MEDIUM

AV:N

AC:L

PR:N

UI:R

S:U

C:H

A:N

Tri-Gauge Impact Visualization

Confidentiality

HIGH

Integrity

NONE

Availability

NONE

Tri-Gauge Impact Vimprints

Confidentiality

HIGH

Integrity

NONE

Availability

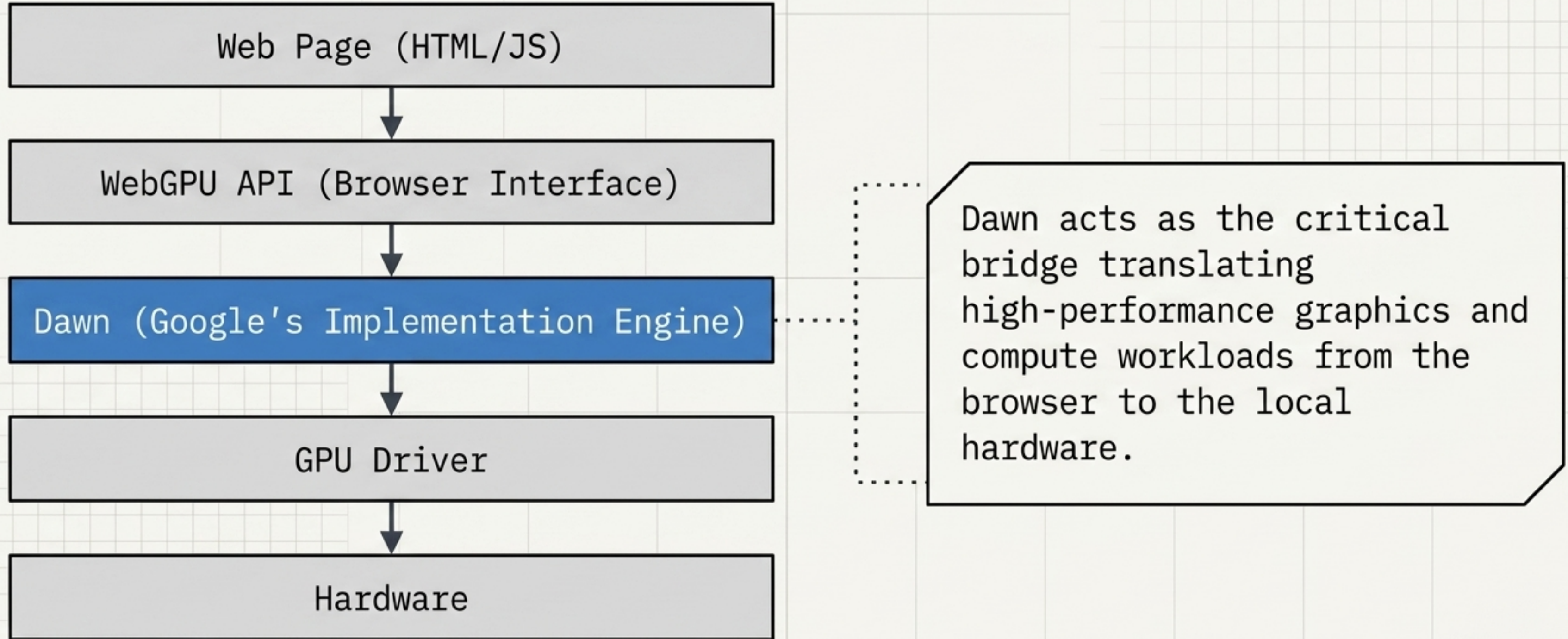
NONE

Diagnostic Quick Facts

|                |                        |
|----------------|------------------------|
| Vendor:        | Google                 |
| Component:     | Dawn (WebGPU)          |
| Vulnerability: | Uninitialized Variable |
| Attack Vector: | Network / Crafted HTML |



# System Context: The WebGPU Architecture





# Core Mechanism: The Memory Initialization Flaw (CWE-457)

## Secure Behavior

```
int value = 0;
```

|    |    |    |    |
|----|----|----|----|
| 00 | 00 | 00 | 00 |
| 00 | 00 | 00 | 00 |
| 00 | 00 | 00 | 00 |
| 00 | 00 | 00 | 00 |

Allocated & Cleared.  
Variable initialized before use.

## Vulnerable Behavior / CWE-457

```
int value;  
printf("%d", value);
```

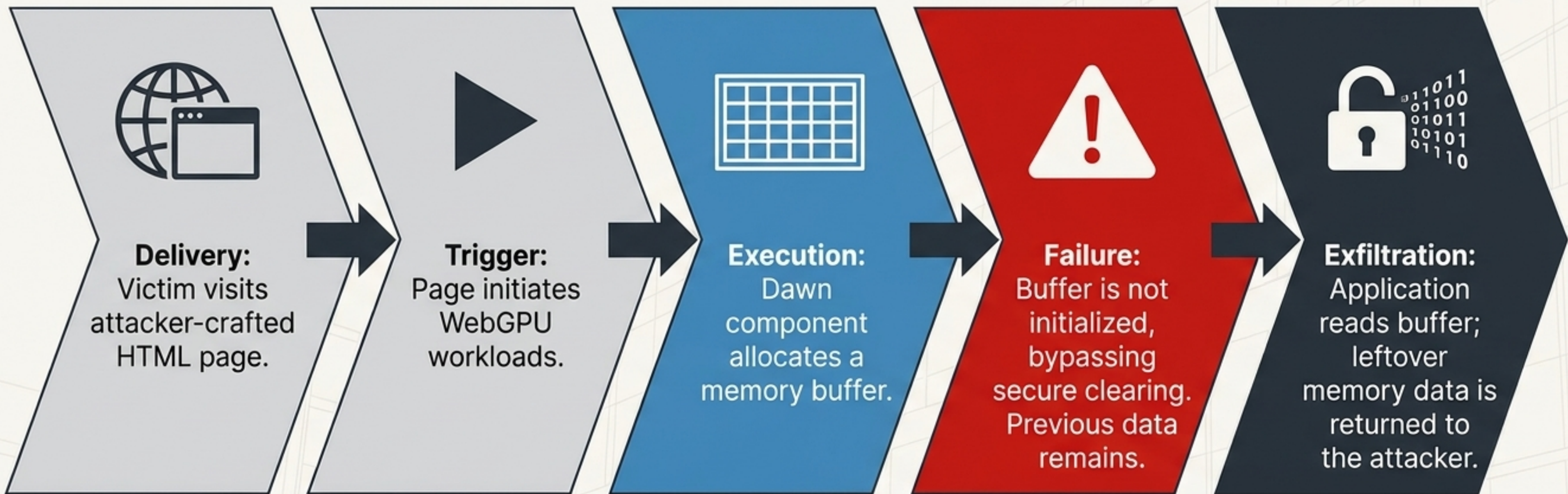
|    |    |       |     |
|----|----|-------|-----|
| A8 | F2 | token | ptr |
| 00 | %A | 0M    | 1B  |
| EA | }, | ^ \   | AZ  |
| U1 | %s | \$.   | 01  |

Allocated, Uninitialized, Exposed.  
Variable used before initialization.

**Root Cause:** CWE-457. The application allocates memory but fails to initialize it with a known value before reading, exposing whatever sensitive data previously occupied that space.



# The Exploit Kill Chain





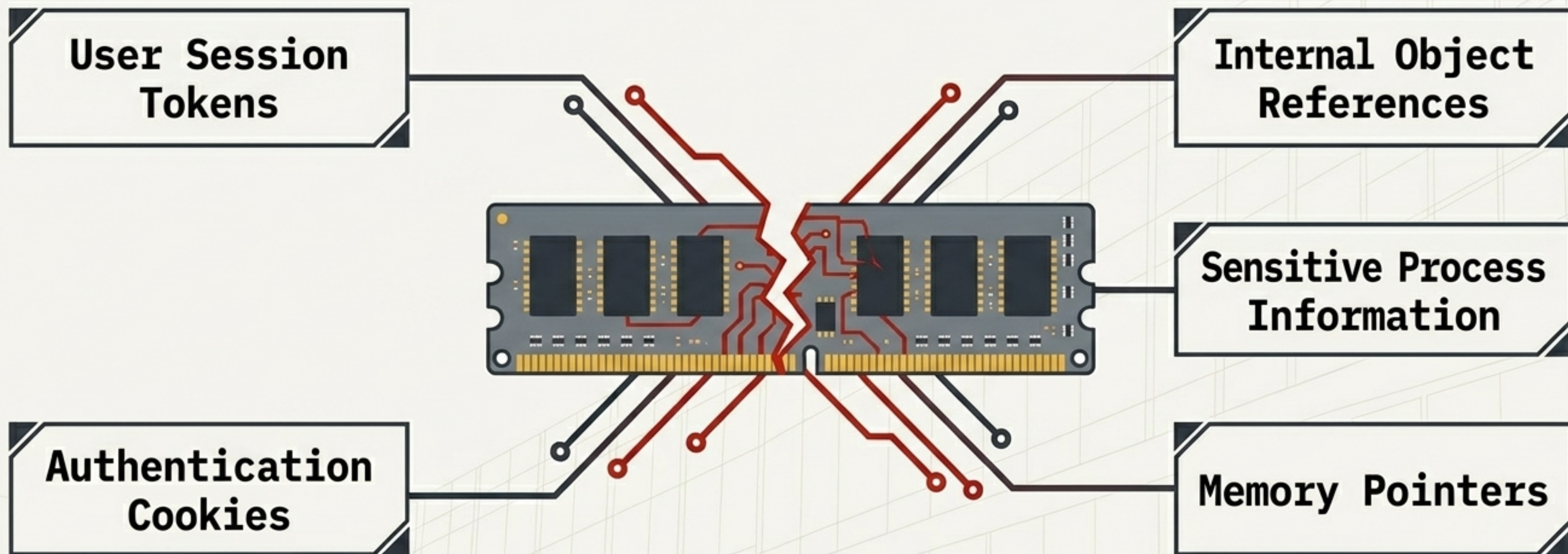
# Attack Requirements & Profiling

| Attack Requirements                                                                                                                 | Not Required                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
|  Network Access                                    |  Authentication / Login      |
|  Crafted HTML Page                                |  Local System Access        |
|  Victim Interaction<br>(Visits the crafted page) |  Administrative Privileges |

**Strategic Takeaway:** This is a zero-click post-navigation exploit. Once a victim lands on the malicious page, the information disclosure triggers silently without requiring any further interaction or credentials.



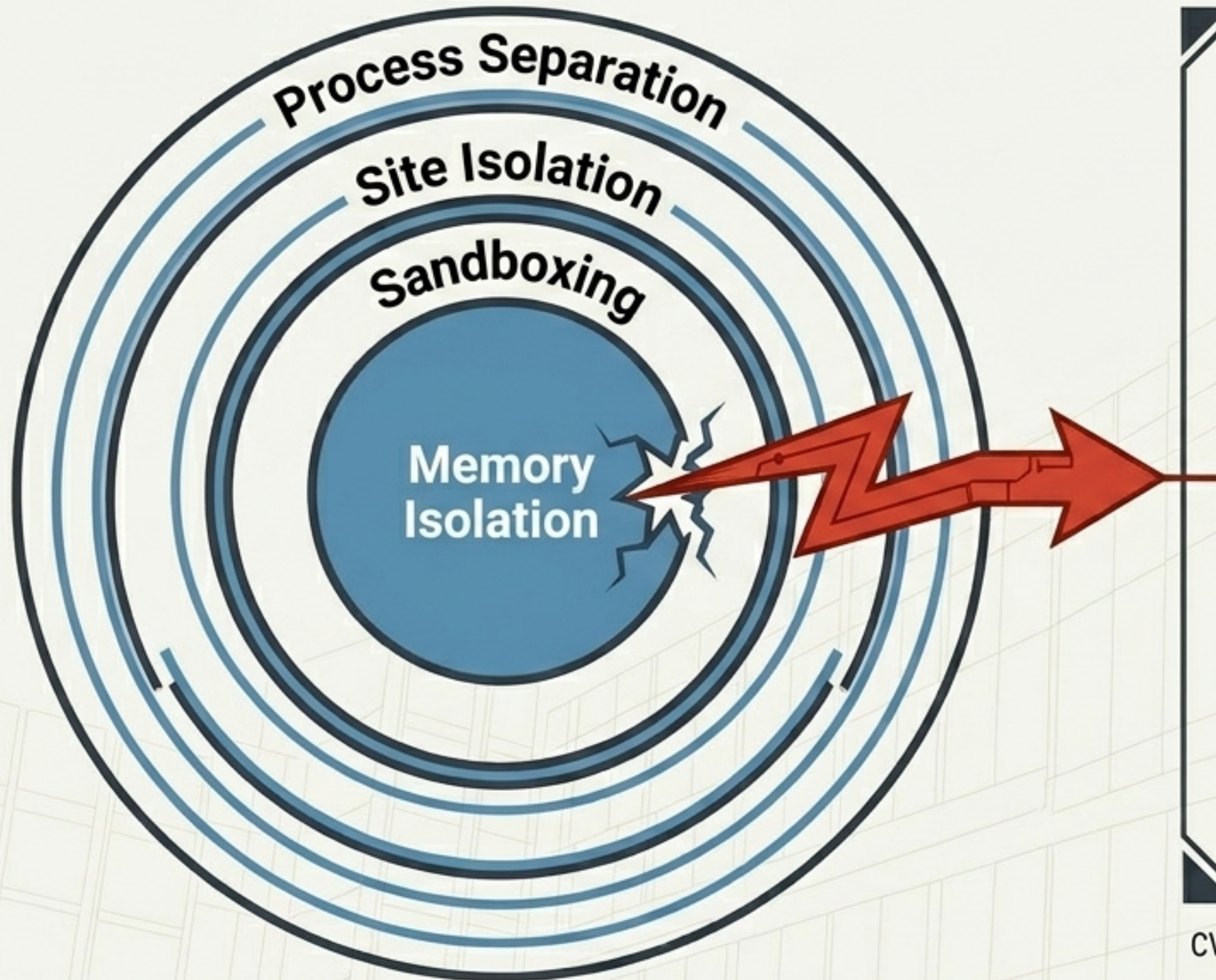
# Payload Analysis: What Exactly is Leaked?



**Insight:** While CVE-2026-11067 does not modify data, leaking this raw memory provides attackers with the exact cryptographic and locational puzzle pieces needed to engineer secondary attacks.



# Strategic Impact on Browser Security



**Modern browsers rely on strict memory boundaries. By exposing internal memory, CVE-2026-11067 weakens these foundational protections, acting as a potential stepping stone for sandbox escapes.**

CVE-2026-11067 Impact Analysis: Memory Boundary Compromise



# Threat Hunting Matrix (MITRE ATT&CK)

|                  |                                                                |                                                                            |
|------------------|----------------------------------------------------------------|----------------------------------------------------------------------------|
| Initial Access   | <div>TECHNIQUE</div> <div>[T1189] Drive-by Compromise</div>    |                                                                            |
| Collection       | <div>TECHNIQUE</div> <div>[T1005] Data from Local System</div> | <div>TECHNIQUE</div> <div>[T1213] Data from Information Repositories</div> |
| Follow-on Impact | <div>TECHNIQUE</div> <div>[T1552] Unsecured Credentials</div>  |                                                                            |

Note: Mapping indicates the primary vectors for initial compromise and the resulting data exposure pathways. Watch for credential harvesting as a secondary objective.



# Telemetry & Detection Opportunities

## Browser Telemetry

-  - Monitor for:  
WebGPU Failures
-  - Monitor for:  
Dawn Component Errors
-  - Monitor for:  
Unexpected Memory Read Spikes
-  - Monitor for:  
Suspicious Web Applications  
calling WebGPU APIs

## EDR / Endpoint Telemetry

-  - Monitor for:  
Chrome Memory Anomalies
-  - Monitor for:  
Suspicious Browser Behavior  
patterns
-  - Monitor for:  
Abnormal GPU Requests from  
browser processes



## Remediation & Mitigation Protocols

# Tier 1: Immediate / Critical Action

UPGRADE CHROME

Target Version: 149.0.7827.53+

## Tier 2: Compensating Controls

**Policy:** Disable unnecessary  
WebGPU usage

**Endpoint:** Increase memory anomaly  
monitoring

**Network:** Enforce strict browser  
isolation protocols

**Operations:** Initiate proactive  
threat hunting for **T1189** indicators



# Executive Conclusion

## 1. The Flaw

Uninitialized memory (CWE-457) in Chrome's Dawn (WebGPU) component allows unauthorized reads of leftover memory data.

## 2. The Risk

Silent, zero-click exposure of sensitive tokens, pointers, and session data triggered merely by visiting a crafted web page.

## 3. The Mandate

Immediate fleet-wide deployment of Chrome version 149.0.7827.53+ is required to restore memory integrity and secure the browser boundary.